

Project Y: First-Person Los Alamos Conversion

Objective

Convert the existing `project_y_15_week_dashboard_v21.html` game into a first-person, explorable Los Alamos town.

The player walks between five scientific buildings, enters laboratories, speaks with division leaders, and completes the existing educational challenges. The game should feel like visiting an active wartime research town rather than navigating a collection of browser panels.

This is an **FPS-style exploration interface**, not a shooting game. There should be no weapons, combat, enemies, or firing mechanics. “FPS-style” refers only to first-person movement, mouse look, environmental interaction, and entering buildings.

The existing HTML contains the curriculum, question bank, leaders, team progress model, money system, weekly simulation, terminology definitions, feedback, and forecasting logic. Preserve those systems. The main task is to replace the flat town-map presentation with an immersive three-dimensional world.

1. Preserve the Existing Game

Treat the current HTML as the authoritative source for:

- All five divisions
- All leaders and leader characteristics
- All 150 curriculum questions
- The 15-week campaign structure
- Random selection of one question from each two-question weekly pair
- Randomized starting readiness centered near 30%
- Team milestone progress
- Team project budgets
- Director funds
- Weekly spending and progress
- Issues and issue risk
- Scientific hints
- Retry mechanics
- Correct-answer feedback
- Wrong-answer feedback that reveals the correct solution
- Clickable terminology definitions
- Current and projected readiness
- Current and projected money
- Local save data

Do not summarize, shorten, regenerate, or replace the existing curriculum. Extract it from the HTML and move it into clean data modules.

Do not create a second independent scoring system for the 3D version. The first-person world must call the same underlying simulation functions used by the current game.

2. Recommended Project Structure

Refactor the single HTML file into a small browser game project:

```
project-y-fps/  
  index.html  
  package.json  
  src/  
    main.js  
    world.js  
    player.js  
    interactions.js  
    gameState.js  
    simulation.js  
    curriculum.js  
    divisions.js  
    leaders.js  
    questions.js  
    dashboard.js  
    questionUI.js  
    terminology.js  
    save.js  
    styles.css  
  public/  
    models/  
    textures/  
    audio/
```

Use:

- Three.js for the three-dimensional town
- `PointerLockControls` for first-person mouse and keyboard movement
- Plain JavaScript modules
- HTML and CSS overlays for question interfaces
- Local storage for campaign saves
- Simple low-poly geometry for the first implementation

The first working version should not depend on custom 3D assets. Construct buildings, roads, signs, laboratory furniture, the central screen, and surrounding terrain from basic geometry. More detailed models can be added later.

3. Core First-Person Controls

Desktop controls:

- **W**, **A**, **S**, **D** : move
- Mouse: look around
- **E** : interact
- **Shift** : walk faster
- **Escape** : release the mouse
- **M** : show a small town map
- **Tab** : show current campaign summary

Place a small crosshair at the center of the screen.

When the player looks at an interactive object within range, show a concise prompt such as:

E – Enter Theory Division

or:

E – View Project Dashboard

Do not leave interaction prompts visible when the player is too far away or looking elsewhere.

Add collision so the player cannot walk through buildings, desks, fences, or the central display.

4. Town Layout

Create a compact town that takes approximately 30–60 seconds to walk across.

The player begins in a central plaza. Five roads or paths lead to the five divisions:

1. **Theory & Calculations**
2. **Experimental Physics & Diagnostics**
3. **Chemistry & Metallurgy**
4. **Ordnance & Engineering**

5. Implosion & Systems Integration

Each building should be visually recognizable from a distance through:

- A large division sign
- A distinct exterior color or accent
- A simple division symbol
- Different equipment visible near the entrance
- A beacon or status light showing the team's condition

Possible environmental identities:

Theory & Calculations

A quieter office and calculation building with blackboards, desks, papers, equations, slide rules, and filing cabinets.

Experimental Physics & Diagnostics

A laboratory with counters, oscilloscopes, cables, detector stands, timing equipment, and test benches.

Chemistry & Metallurgy

A laboratory and workshop with furnaces, sample containers, ventilation equipment, microscopes, molds, and materials-testing stations.

Ordinance & Engineering

An engineering shop with mechanical drawings, casings, aircraft diagrams, tool benches, test fixtures, and component mockups.

Implosion & Systems Integration

A test and coordination building with lens diagrams, timing circuits, radiography images, diagnostic layouts, and system-integration boards.

Keep the environment museum-appropriate. Do not depict operational bomb assembly instructions, realistic component dimensions, or usable weapons diagrams.

5. The Central Project Screen

Place a large, highly visible electronic or mechanical status board in the center of town.

It should be possible to understand the broad state of the project while standing several meters away.

The board must show one row for each division:

- Division name
- Current productivity/readiness
- Projected readiness when Week 15 ends
- Current project budget
- Projected project budget when Week 15 ends
- Current issue status

Team projections must never exceed 100%.

At the top of the board, show:

- Current week: `Week X of 15`
- Overall readiness now
- Projected overall readiness
- Director funds now
- Projected director funds
- Number of weeks remaining

Use two bars for each team:

- Solid bar: current readiness
- Lighter or outlined extension: projected Week 15 readiness

Use the same visual system for money:

- Current project money
- Projected money remaining

The board must update immediately after:

- Completing a visit
- Answering correctly
- Answering incorrectly
- Funding a team
- Buying a hint
- Buying a retry
- Advancing the week
- A team completing a milestone
- A new issue appearing

When the player approaches the board and presses `E`, open an expanded dashboard overlay with the complete information already calculated by the existing `renderDashboard()` and forecasting logic.

The central board should not contain long explanatory paragraphs. It is a status display.

6. Building Status in the World

Each building should communicate its condition without requiring the player to open a menu.

Use a status fixture beside the door:

- Green: progressing normally
- Amber: falling behind projection
- Red: active issue
- Blue or white: completed
- Dim: insufficient project budget

A small exterior board can display:

```
THEORY DIVISION
Readiness: 42%
Projected: 88%
Budget: $39
```

Do not show the question title, lesson topic, game type, or long status sentences outside the building.

7. Entering a Building

When the player presses **E** at a door:

1. Fade briefly to black.
2. Place the player inside a compact laboratory or office.
3. Position the division leader near an interactive desk, board, or instrument.
4. Let the player walk around the room.
5. Show **E – Speak with Hans Bethe** or the appropriate leader.
6. Starting the conversation opens the current week's educational challenge.

The leader should be represented initially by a simple low-poly person, portrait board, or stylized character. Do not block development on realistic human models.

Each building only needs one strong interior room for the first version.

8. Educational Challenge Presentation

Keep the existing four challenge formats, but present them as activities inside the laboratory.

When a challenge begins, pause player movement and unlock the mouse so the player can interact with the educational interface.

Protocol

Show the scientific situations on a laboratory board.

The player matches each situation to a concrete action, measurement, conclusion, or response.

Remove redundant headings such as:

- "Choose the team's response"
- "Situation written on the board"
- "Player task"
- "Setup"

Show only:

- The science-story paragraph
- The situations
- The answer choices
- A Check button

Sequence

Place four scientific steps as cards, documents, photographs, or procedure tiles on a desk.

The player arranges them in order.

The answer choices should retain their complete educational wording rather than becoming short category labels.

Ballpark

Show four candidate numerical estimates as notes or cards on a calculation board.

The problem must remain a genuine Fermi-estimation challenge:

- Do not state the required values numerically in the prompt.
- The player chooses plausible estimates.
- The player places two or three estimates into the displayed relationship.
- The game calculates and checks the order-of-magnitude result.

Science Tank

Represent research proposals as folders, equipment requests, or project cards.

The player allocates limited resources among competing proposals.

Each proposal must explain what scientific work the investment supports.

9. Science Story Requirements

Every question begins with one concise paragraph that:

1. Introduces the larger scientific or engineering idea.
2. Explains the physical problem in language a high-school student can follow.
3. Shows how Los Alamos encounters that problem while developing the bomb.
4. Ends with the decision, prediction, ordering problem, or estimate facing the team.

The paragraph must teach the science, not merely describe the game.

Avoid language such as:

- "This project milestone requires..."
- "The player must..."
- "The team's board shows..."
- "A correct intervention adds..."
- "This question tests..."

The connection to bomb development must be part of the scientific story rather than a separate "why this matters" box.

10. Terminology

Retain the existing clickable terminology system.

Technical terms appearing in the science story, question, answer choices, or feedback should be visually marked.

Clicking a term opens a concise definition.

The definition must be available in every question where the term appears. Do not assume the player opened or remembered an earlier definition.

In the 3D presentation, terminology can appear as underlined words, small term buttons, or glossary chips in the question overlay.

11. Answer Feedback

Correct-answer feedback should show:

- The correct answer
- The scientific reasoning
- What the evidence or calculation demonstrates
- How the result affects understanding of the larger system

Wrong-answer feedback must show:

- What the player submitted
- The complete correct answer
- Why the submitted answer is not supported
- The scientific reasoning behind the correct answer
- Definitions of important terms
- The connection to the larger bomb-development problem

Do not hide the correct answer after an incorrect attempt.

Keep the feedback focused on science and engineering rather than readiness points.

After the player closes the feedback, return them to the building interior.

12. Weekly Campaign Loop

Preserve the existing 15-week structure.

Each campaign week follows this loop:

1. The player begins in the central plaza.
2. The central screen shows current and projected status.
3. The player reviews the five buildings.
4. The player chooses one division to visit.
5. The player completes that division's current challenge.
6. The player may spend director funds.
7. The player returns to the central screen.
8. The player advances the calendar.
9. Automatic team work, spending, milestones, issues, and weekly funding are processed.
10. The town and central screen update for the new week.

The existing game permits one formal division visit per week. Preserve this rule.

After visiting one division, the other buildings may still be entered for observation, but their formal challenge should be unavailable until the next week. Show a simple message such as:

This week's formal division visit is complete.

13. Money

Preserve the simplified money system:

- Start with \$30 in director funds.
- Receive \$5 at the start of each new week.
- Spend \$1 to add 1% readiness to a selected team.
- Spend \$2 for a scientific hint during a challenge.
- Spend \$3 for one retry after an incorrect answer.

Allow team funding from either:

- The central project screen
- A funding terminal inside the selected team's building

For direct funding, provide:

- Fund \$1
- Fund \$5, when affordable
- Use all available funds

Always show the exact result before confirmation:

Spend \$5 to raise Chemistry & Metallurgy from 46% to 51%?

Cap readiness at 100%.

14. Initial Campaign Setup

Retain the leader-assignment phase.

It can remain a conventional overlay before the player enters the town.

After the player assigns leaders:

- Generate randomized starting readiness using the existing formula.
- Keep the median near 30%.
- Preserve team difficulty and leader science/management effects.
- Generate the 15-week randomized lesson schedule.

- Initialize remaining project budgets.
- Create any starting issues.
- Spawn the player in the central plaza.

The central board should immediately reveal that the five teams begin at different levels.

15. Saving

Continue saving the campaign locally.

Save at least:

- Week
- Leader assignments
- Team readiness and milestones
- Remaining project budgets
- Director funds
- Selected weekly questions
- Completed visit
- Hints
- Retries
- Issues
- Logs
- Player position, optionally

Attempt to recognize the existing local-storage key:

```
projectY_15week_dashboard_v21
```

Either migrate that save into the new format or provide a one-time “Import existing campaign” option.

Do not silently delete an existing campaign.

16. Visual Direction

Use a stylized, low-poly, museum-exhibit aesthetic rather than photorealism.

Desired qualities:

- Warm desert light
- Mountain backdrop
- Dusty roads

- Timber and utilitarian laboratory buildings
- Hand-painted division signs
- Scientific equipment visible through windows
- Period-inspired typography
- Clear modern educational interfaces
- Strong contrast and readable text

The town should feel inhabited and purposeful, but it does not need crowds.

Simple ambient activity can include:

- Scientists walking between buildings
- Trucks parked beside laboratories
- Papers or equipment being carried
- Distant machinery sounds
- Lights changing as teams encounter issues or complete milestones

Ambient characters do not need dialogue in the first version.

17. Accessibility and Fallbacks

Include:

- Mouse sensitivity setting
- Movement-speed setting
- Reduced-motion option
- Volume controls
- Captions for spoken or ambient information
- High-contrast question interface
- Keyboard-accessible question controls
- A button to exit first-person mouse capture
- A conventional 2D campaign dashboard fallback

The educational questions must remain usable without precise first-person movement.

On mobile or unsupported devices, load the existing 2D interface rather than attempting full first-person controls.

18. Implementation Order

Phase 1: Preserve the simulation

Extract the curriculum, divisions, leaders, game state, forecasts, question checking, terminology, and weekly simulation from the existing HTML.

Add automated checks confirming:

- Five divisions exist.
- Each division contains 30 lessons.
- The campaign contains 15 weeks.
- Each weekly lesson comes from the correct pair.
- Starting readiness remains within the intended range.
- Readiness never exceeds 100%.
- Wrong answers reveal the correct solution.
- Funding, hints, and retries charge the correct amounts.
- Forecast values update after state changes.

Phase 2: Build the first-person town

Create:

- Central plaza
- Five exterior buildings
- Roads and terrain
- Player movement
- Collision
- Interaction raycasting
- Central project screen
- Building entrances

Phase 3: Build one complete division

Implement Theory & Calculations from start to finish:

- Exterior
- Interior
- Leader
- Question interaction
- Terminology
- Feedback
- Funding
- Return to town
- Dashboard update

Use this as the template for the remaining four divisions.

Phase 4: Add all divisions

Connect the remaining question banks without changing their content or logic.

Phase 5: Campaign polish

Add:

- Week transitions
- Status lights
- Issue indicators
- End-of-campaign sequence
- Sound
- Accessibility
- Save migration
- Performance optimization

19. Acceptance Criteria

The conversion is complete when:

1. The player can walk around a three-dimensional Los Alamos town.
2. All five divisions are physically represented.
3. The player can enter every building.
4. Each building serves the correct division's weekly question.
5. All existing question formats work.
6. Clickable terminology works.
7. Wrong answers reveal the correct answer and reasoning.
8. The existing progress and money simulation remains functional.
9. The central town screen shows current and projected readiness for all teams.
10. The central town screen shows current and projected money.
11. Dashboard values update immediately after decisions.
12. The campaign lasts 15 weeks.
13. Only one formal division visit is allowed per week.
14. Save and reload work.
15. No question-bank content is lost.
16. There are no combat or shooting mechanics.
17. The experience remains understandable to a high-school student.
18. A non-3D fallback remains available.

20. First Deliverable

Do not attempt the entire polished game in the first run.

Produce a playable vertical slice containing:

- The central plaza
- The large project dashboard
- All five building exteriors
- Working first-person movement
- Working building interaction
- A complete Theory & Calculations interior
- One Protocol challenge
- One Sequence challenge
- One Ballpark challenge
- One Science Tank challenge
- Correct and incorrect feedback
- A working terminology pop-up
- Direct funding
- Week advancement
- Save and reload

Use placeholder low-poly art. The priority is proving that the existing educational and project-management systems can operate naturally inside the first-person world.

After the vertical slice is stable, connect the complete 150-question curriculum.